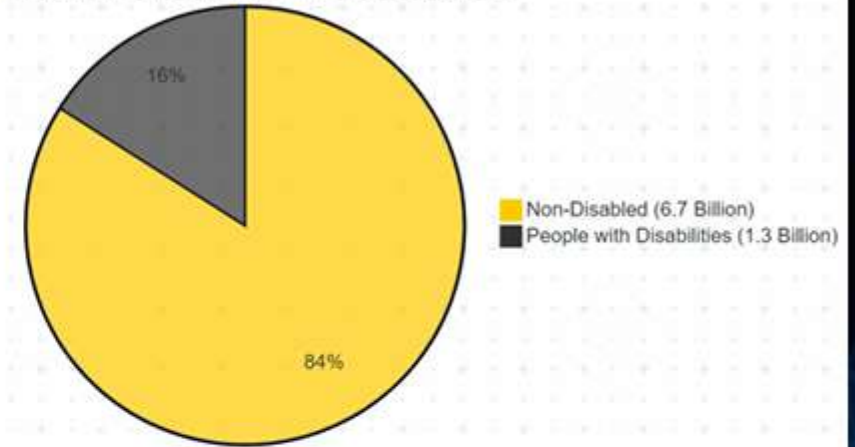


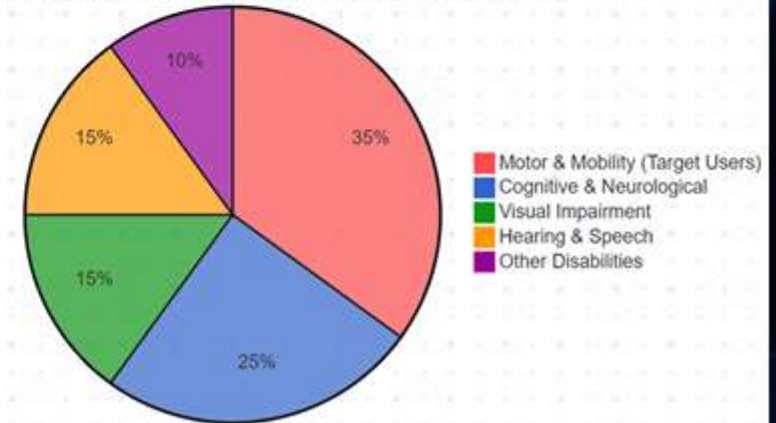
# IDEA TITLE - De-disabled

- Across the globe, over 1.3 billion people live with significant disabilities, yet millions are effectively locked out of the digital world because they cannot use standard interfaces like keyboards or touchscreens.
- Financial Barrier: Essential eye-tracking hardware costs \$2,500+ per unit, making it unaffordable for 90% of the target population.
- Zero Autonomy: Non-verbal users currently have 0% independent means to cast votes or make decisions without caretaker assistance.
- Passive Failure: Traditional, non-gamified therapy tools suffer from <15% user retention, leading to slow cognitive recovery.

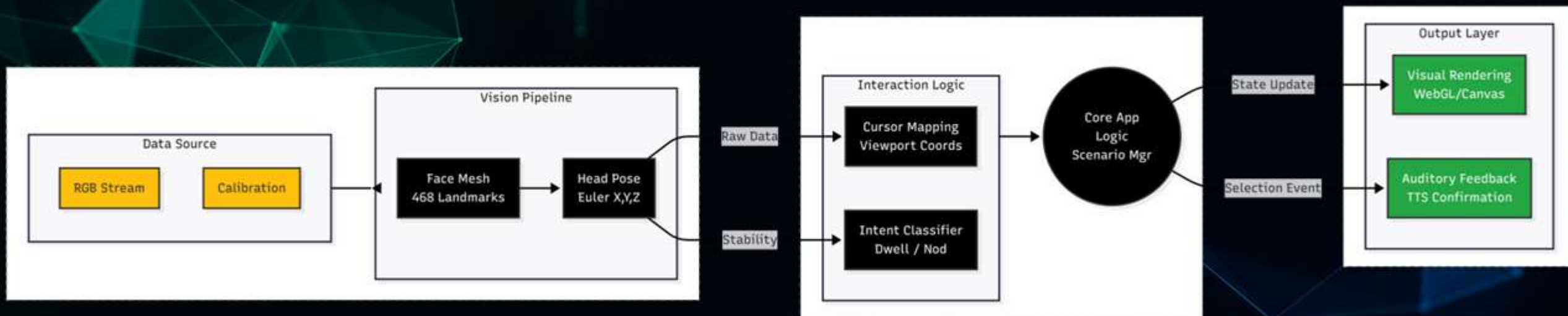
GLOBAL POPULATION (16% Disabled)



TYPES OF DISABILITY (Breakdown of the 1.3 Billion)



# TECHNICAL APPROACH



# FEASIBILITY AND VIABILITY

- **Analysis of Feasibility**

- Technical: Built on standard Web Technologies (PWA, WebGL), ensuring compatibility with 90% of devices (standard laptops) without GPU requirements.
- Economic: ₹0 Hardware cost. Leverages the user's existing webcam, removing the ₹2,00,000+ barrier to entry (cost of imported eye-trackers).
- Challenges & Risks
  - Environmental: Poor lighting or low-resolution cameras may affect tracking accuracy.
  - Physiological: Severe tremors or spasms could trigger "false positive" clicks.
- Mitigation Strategies
  - Adaptive AI: "Intent Classifier" algorithm filters out micro-jitters/spasms.
  - Dynamic Calibration: 5-second setup adjusts sensitivity to the user's range of motion.

# IMPACT AND BENEFITS

- 1. Potential impact on the target audience
  - Restored Autonomy: Grants non-verbal users (ALS, Cerebral Palsy) 100% independent power to cast votes and make decisions without caretaker assistance.
  - "Stealth" Therapy: Transforms repetitive rehabilitation into fun gaming; users unknowingly strengthen neck muscles and improve motor coordination while playing.
  -
- 2. Benefits of the solution (Social, Economic, Environmental)
  - Economic Benefit: Zero Hardware Cost. Replaces imported eye-trackers worth ₹2,00,000+ with free software that runs on existing webcams.
  - Social Benefit: Directly targets SDG 10 (Reduced Inequalities) and SDG 4 (Quality Education) by bringing assistive tech to rural and low-income schools.

# RESEARCH AND REFERENCES

- **Technical References & Libraries**
- **Google MediaPipe Face Mesh:** Used for high-fidelity 468-point facial landmark detection. (Source: [developers.google.com/mediapipe](https://developers.google.com/mediapipe))
- **TensorFlow.js:** Utilized for client-side Machine Learning inference to ensure zero-latency privacy. (Source: [tensorflow.org/js](https://tensorflow.org/js))
- **WCAG 2.1 Guidelines:** Interaction design aligns with Web Content Accessibility Guidelines for motor impairments. (Source: [w3.org/WAI](https://w3.org/WAI))
- **Market & Statistical Data**
- **World Health Organization (WHO):** Data regarding the 1.3 billion disabled population and the "Assistive Technology Gap." (Source: WHO Global Report on Health Equity for Persons with Disabilities, 2022)
- **Inspiration & Benchmarking**
- **Project Euphonia (Google):** Research on AI for non-standard speech, which inspired our non-standard gesture approach.